

N100 Financial Intelligence Platform

Analyst Guide

Sprint 6 — API Server, Clustering & Final QA

Version 1.0

This guide explains how to use the N100 Financial Intelligence Platform, including the Streamlit dashboard, screener, company analysis, PDF tearsheets, REST API, clustering outputs, and troubleshooting procedures.

1. Platform Overview

The N100 Financial Intelligence Platform combines financial data engineering, financial ratio analysis, CAGR analysis, cash-flow analysis, clustering, visual dashboards, PDF reports, and a FastAPI service.

Primary data sources

- companies.xlsx — company master information, ROCE and ROE.
- market_cap.xlsx — market capitalisation and valuation history.
- financial_ratios.xlsx — historical financial ratios.
- profitandloss.xlsx — historical P&L; information.
- balancesheet.xlsx — historical balance-sheet information.
- cashflow.xlsx — operating, investing and financing cash flows.
- sectors.xlsx — broad sector, sub-sector and index information.
- peer_groups.xlsx — peer-group membership and benchmarks.

The platform currently covers 92 companies in the company master dataset.

2. Running the Streamlit Dashboard

Start the dashboard from the project root using:

```
streamlit run dashboard/app.py
```

The dashboard is designed as an analyst-facing interface for exploring company information, financial metrics, valuation data, peer information, and analytical outputs.

Typical workflow

- Open the Streamlit application in the browser.
- Select the required dashboard page.
- Choose a company or apply available filters.
- Review the displayed financial metrics and analytical results.
- Use the screener to narrow the company universe.
- Use company-level analysis for detailed investigation.

The dashboard should be started from the project environment so that relative paths to data and output files resolve correctly.

3. Using the Financial Screener

The screener filters companies using financial metrics from the engineered financial-ratio dataset.

Available filters

- Minimum ROE — filters companies at or above the selected ROE.
- Maximum Debt-to-Equity — filters companies at or below the selected leverage.
- Minimum Operating Profit Margin — filters companies above the selected OPM.
- Minimum PE — filters companies at or above the selected PE.
- Maximum PE — filters companies at or below the selected PE.
- Result limit — controls the maximum number of returned records.

Example API screener request:

```
GET /api/v1/screener/?min_roe=40
```

The screener output is also written to output/screener_output.csv when the API endpoint is called.

4. Company Profile Analysis

The company profile endpoint provides company-level information together with the latest available analytical metrics.

Example:

GET /api/v1/companies/TCS

Related company endpoints

- GET /api/v1/companies/{ticker}/pl — P&L; history.
- GET /api/v1/companies/{ticker}/bs — balance-sheet history.
- GET /api/v1/companies/{ticker}/cashflow — cash-flow history.
- GET /api/v1/companies/{ticker}/ratios — computed financial ratios.
- GET /api/v1/companies/{ticker}/documents — company document links.
- GET /api/v1/companies/{ticker}/tearsheet — company PDF tearsheet.
- GET /api/v1/companies/{ticker}/peers/compare — peer comparison data.

5. Valuation and Market-Cap Analysis

The valuation module provides company-level valuation information and historical market-cap multiples.

Valuation endpoint

GET /api/v1/valuation/{ticker}

Historical market-cap endpoint

GET /api/v1/market-cap/{ticker}

The historical market-cap data covers the available 2019–2024 valuation history in the source dataset.

Typical valuation fields include P/E, P/B, EV/EBITDA, dividend yield, market capitalisation and enterprise value.

Analysts should interpret valuation metrics together with profitability, leverage, growth and cash-flow information rather than relying on one metric alone.

6. Peer and Sector Analysis

Peer groups are defined in peer_groups.xlsx. The platform contains 11 peer groups.

Peer groups

- Private Banks
- Public Sector Banks
- IT Services
- Pharmaceuticals
- Automobiles
- Life Insurance
- Oil & Gas
- Power & Utilities
- Steel
- FMCG
- Consumer Finance

Example endpoint:

GET /api/v1/peers/Private%20Banks

7. Clustering and Portfolio Analytics

The clustering module assigns the 92 companies to five KMeans clusters.

Clustering features

- Return on equity percentage.
- Debt-to-equity ratio.
- Five-year revenue CAGR.
- Five-year free-cash-flow CAGR.
- Operating profit margin percentage.

The clustering workflow uses sector-median imputation for missing feature values, StandardScaler and KMeans with five clusters and random_state=42.

Key outputs

- output/cluster_labels.csv
- output/cluster_profiles.csv
- reports/elbow_plot.png
- output/portfolio_stats.csv

8. Generating and Using PDF Tearsheets

Company tearsheets are generated as PDF reports and stored under `output/reports/tearsheets/`.

Typical generation workflow

- Run the tearsheet generation module from the project environment.
- Verify that PDF files are created under `output/reports/tearsheets/`.
- Check PDF page counts and file sizes.
- Open representative PDFs to review layout and readability.

The API also exposes a company tearsheet endpoint:

`GET /api/v1/companies/{ticker}/tearsheet`

When reviewing reports, check for missing companies, unexpected blank pages, clipped text and chart overflow.

9. REST API Quick Reference

Start the API with:

```
uvicorn src.api.main:app --reload
```

Health check

```
GET /api/v1/health/
```

Company list

```
GET /api/v1/companies/
```

Screener

```
GET /api/v1/screener/?min_roe=40
```

Portfolio statistics

```
GET /api/v1/portfolio/stats
```

Portfolio clusters

```
GET /api/v1/portfolio/clusters
```

The interactive Swagger documentation is available at:

```
http://127.0.0.1:8000/docs
```

10. Troubleshooting

API connection refused

If requests return WinError 10061, confirm that Uvicorn is running on port 8000 and keep its terminal open.

Missing output file

Check that the required source data and output directories exist. Run the relevant analytics module before calling the dependent API endpoint.

Import or module error

Run commands from the N100 Financial Intelligence Platform project root and confirm the Python environment contains the required packages.

PDF generation issue

Check that ReportLab is installed and that the output directory is writable.

Pytest failure

Run pytest from the project root and inspect the failing test before changing application code.

11. Quality Assurance Checklist

Before final delivery, verify the following:

- 92 companies are present in the company master data.
- At least 90% of companies have the required historical records.
- SQLite foreign-key validation returns zero violations.
- Financial-ratio row count meets the Sprint 6 threshold.
- Revenue CAGR spot-check is within the specified tolerance.
- ROE validation has five companies within the required tolerance.
- The quality screener returns the required range.
- Company profile response time is below the specified limit.
- Screener CSV is generated and valid.
- Tearsheet layout has been reviewed.
- Health endpoint returns HTTP 200.
- TCS ratio history contains at least ten years.
- Peer-group coverage is verified.
- All companies have cluster assignments.
- Pros/cons coverage is reviewed for all companies.
- Tearsheet count and file sizes are checked.
- Pytest has at least 60 tests with zero failures.
- Validation failures are documented.
- Analyst guide exists and contains at least ten pages.

End of Analyst Guide